
Distributions, Regression, and Inference — Diagnosing Causal Claims and Biased Designs

Every claim in this set comes with a number that is arithmetically correct and a conclusion that is not. Your job is to compute the number the claim rests on and then say exactly what the design will and will not support.

- **Coverage** — who could ever have been selected?
- **Response** — of those reached, who answered, and are they special?
- **Assignment** — who decided which subjects got the treatment?
- **Attrition** — who left the study before it ended, and were they dropped from the arithmetic?

Association is what an observational study can establish. Causation needs randomized assignment.

1. The Ridgeway Cycling Club has 400 members. It posted a survey link on its website, and 96 members answered. The club's newsletter reports the result as "what the club thinks". What fraction of the membership answered the survey? Give the exact fraction in lowest terms.

Answer: _____

2. Diagnose the design. The newsletter treats the answers as the view of the whole membership. Name the two distinct biases at work when a survey is posted where members choose whether to answer, and describe which kind of member is over-represented among the people who did answer.

3. Find and fix. A club officer says, “The coverage is fine — almost a quarter of the members were covered,” and writes 0.24. That number is the response rate, not the coverage. In fact only 260 of the 400 members ever visit the club website, so the rest could never have seen the link at all. What is the probability that a member chosen at random from the membership list was even reachable by this survey? Give the exact fraction.

Answer: _____

4. The Lighthouse Wellness program enrolled seven volunteers. The table gives each one’s weight change over twelve weeks, in kilograms lost (a negative entry means the person gained). The last two enrollees quit before the end.

9	3	12	5	1	−2	0
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The company’s advertisement reports a mean loss of 6 kilograms, computed from the finishers only. Compute the mean weight change over *everyone who enrolled*.

Answer: _____

5. The comparison group. Seven members of the same gym who did not enroll in the program were followed for the same twelve weeks. Their weight changes, in kilograms lost, were:

3	0	5	2	-1	2	3
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Compute the mean weight change for this group.

Answer: _____

6. Diagnose the claim. Using the two means you just computed, the company advertises: “Our program makes you lose twice as much weight.” Explain why these data cannot support that causal claim. Your answer must name (a) how subjects came to be in the program group, (b) one confounding variable that plausibly differs between the groups, and (c) the study design that *would* license a causal conclusion, including how subjects are assigned.

7. Find and fix. A pollster interviewed shoppers at a weekday-morning farmers' market and reported that 55% of the sample was retired, while the town itself is 20% retired.

Retired share of the sample	55%
Retired share of the town	20%

The pollster writes “retirees are over-represented by 2.75” after dividing the two percentages. That is a ratio, not a gap. By how many percentage points are retirees over-represented in this sample?

Answer: _____

8. Challenge. The pollster's report is headlined “The town opposes the new library levy.” Rewrite the conclusion so that it claims only what this sample can support. Then design the sampling plan that would license the headline as written: say what the sampling frame is, how respondents are chosen, and what has to be done about the people who do not answer.